

- 1 A random sample of 12 observations of a normal random variable is taken. The results give unbiased estimates for the population mean and variance as 10.24 and 0.52 respectively.

Test, at the 10% significance level, the null hypothesis that the population mean is 10.6 against the alternative hypothesis that the population mean is less than 10.6. [4]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.